

Nikita Kazeev

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Research Scientist at the intersection of AI and Physics

Work Experience

National University of Singapore

2022–present

Postdoc under Kostya Novoselov

Transformer for generating symmetric crystals [ICML 2025](#) · [code](#)

- Created a generative model for material design based on symmetry inductive bias.
- Implemented the model in PyTorch.
- Achieved the best generation quality and diversity among space-group-conditioned models.
- Led a team of 6.

ML for predicting properties of defects in 2D crystals [paper 1](#) · [paper 2](#) · [patent](#) · [conference](#) · [code](#)

- Created a sparse representation of crystals with defects, achieving 4x energy prediction error reduction.
- Developed code for reproducible parallel running of machine learning experiments.
- Led a team of 3.

Constructor Group

February 2021–present

Machine Learning Consultant

AI/ML Consulting

CERN [Yandex → HSE University]

2014–2022

Researcher

[2025 Breakthrough Prize in Fundamental Physics](#) as a member of LHCb.

Generative models for fast simulation [paper 1](#) · [paper 2](#) · [paper 3](#) · [code](#)

- Developed a GAN-based machine learning model for high-fidelity simulation of a Cherenkov detector trained on tabular data.
- Achieved approximately 10^5 speed-up compared to the ab-initio simulator.

Machine learning on noisy data [paper 1](#) · [paper 2](#) · [code](#)

- Developed a rigorous way to train ML algorithms on data with the label-noise model common in high-energy physics.

Muon identification at the LHCb experiment at CERN [paper](#) · [poster](#) · [IDAO-2019](#)

- Solved a classification problem over tabular data with noisy labels under timing constraints.
- Developed a gradient boosting model that reduced the false positive rate by 30% in the critical low-track-momentum region.
- Integrated the model into LHCb production, mostly in C++.
- Packaged the work as a data science competition problem for IDAO-2019.

Education

Higher School of Economics (HSE University) **2016–2020**

PhD in Computer Science, supervisor [Andrey Ustyuzhanin](#); [Machine Learning for particle identification in the LHCb detector](#).

Sapienza — Università di Roma **2016–2020**

PhD in Physics (double degree with HSE), supervisor [Barbara Sciascia](#).

Product Management course at Yandex **Spring 2018**

Focused coursework in product strategy and execution.

Yandex School of Data Analysis **2013–2015**

Master's level CS course covering algorithms, machine learning, deep learning, and distributed systems.

Moscow Institute of Physics and Technology

MS in Physics (2014–2016): Optimisation of data processing of the LHCb experiment.

BS in Physics (2010–2014): Study of the quantum states of the electrons in nonideal plasma and selected molecules using wave packet molecular dynamics with packet splitting.

Mentorship

- Mentored 9 students ([1](#), [2](#), [3](#), [4](#), [5](#), [6](#), [7](#), [8](#), [9](#)), 3 interns, and student workshop projects ([1](#), [2](#)).
- Co-PI of a \$3.4m AI Singapore grant on multiscale machine learning.

Teaching & Outreach

- Pitch research to the general public through major newspaper coverage ([1](#), [2](#)), blog posts ([1](#), [2](#)), a [science slam](#), interviews ([1](#), [2](#)), and a public [talk](#).
- Taught at the Machine Learning in High Energy Physics Summer School in [2015](#), [2016](#), [2017](#), [2018](#), [2019](#), [2020](#), [2021](#), and [2023](#).
- Lecturer in the [Coursera Advanced Machine Learning specialisation](#), which is [award-winning](#).
- Delivered more than 20 [conference talks](#).

Service

- Reviewer for RSC Advances, Machine Learning: Science and Technology, and AI for Accelerated Materials Design workshops at NeurIPS and ICLR (2023–2025).
- Peer Staff Supporter at NUS, serving as a first-line support for mental wellbeing.

Other

- Aspiring stand-up comedian; bombed more than 20 times at open mics.
- International Junior Science Olympiad 2008 in Korea, silver medal.

Skills

- **ML & AI** — Structured, non-text, domain-specific data
- **Python** — Research coding
- **PyTorch** — Deep learning
- **C++** — Production
- **Linux** — Administration
- **Research Writing & Speaking**
- **Physics** — Deep domain understanding
- **Leadership & Mentorship**

Selected Publications

1. **Kazeev, Nikita**, and Ian Babich. [Foresight-Phys: A Benchmark for Forecasting the Results of Physical Experiments](#). OpenReview, June 2026.
2. Artem Dembitskiy, Shuya Yamazaki, Artem Maevskiy, **Nikita Kazeev**, Roman A. Eremin, Semen Budenny, Kedar Hippalgaonkar, Antonio Helio Castro Neto, and Andrey E. Ustyuzhanin. [Generative Pipeline for Discovering Solid-State Battery Materials with Universal Atomistic Potentials](#). ICML 2026 AI for Science Workshop, May 2026.
3. Nian Liu, **Nikita Kazeev**, Stephen Gregory Dale, Artem Maevskiy, Yuwei Zeng, Ryoji Kubo, Pengru Huang, Thomas Laurent, Yann LeCun, Kostya S. Novoselov, and Xavier Bresson. [Crys-JEPA: Accelerating Crystal Discovery via Embedding Screening and Generative Refinement](#). arXiv preprint arXiv:2605.14759, May 2026.
4. Nian Liu, Yuwei Zeng, Ryoji Kubo, **Nikita Kazeev**, Stephen Gregory Dale, Artem Maevskiy, Pengru Huang, Thomas Laurent, Kostya S. Novoselov, and Xavier Bresson. [Composable Crystals: Controllable Materials Discovery via Concept Learning](#). arXiv preprint arXiv:2605.14769, May 2026.
5. Jose Recatala-Gomez, **Nikita Kazeev**, et al. [Generative design of inorganic materials](#). arXiv preprint arXiv:2604.14082, April 2026.
6. Alexandre Duval, Siddharth Betala, Samuel P. Gleason, Andy Xu, Georgia Channing, Daniel Levy, Ali Ramlaoui, Clémentine Fourrier, Chaitanya K. Joshi, **Nikita Kazeev**, Sékou-Oumar Kaba, Félix Therrien, Alex Hernández-García, Rocío Mercado, N M Anoop Krishnan. [LeMat-GenBench: Bridging the gap between crystal generation and materials discovery](#). AI for Accelerated Materials Design - NeurIPS 2025 / arXiv preprint arXiv:2512.04562, October 2025.
7. Andrey Okhotin, Maksim Nakhodnov, **Nikita Kazeev**, Mikhail Lazarev, Andrey E. Ustyuzhanin, and Dmitry Vetrov. [MiAD: Mirage Atom Diffusion for De Novo Crystal Generation](#). arXiv preprint arXiv:2511.14426, November 2025.
8. Stephen G. Dale, **Nikita Kazeev**, et al. [AI4X Roadmap: Artificial Intelligence for the advancement of scientific pursuit and its future directions](#). arXiv preprint arXiv:2511.20976, November 2025.
9. M. Y. Lukianov, A. Maevskiy, **N. Kazeev**, D. Mylnikov, D. A. Svintsov, K. S. Novoselov, A. Ustyuzhanin, and D. A. Bandurin. [Inverse design of broadband antennas for terahertz devices based on two-dimensional materials](#). *Physical Review Applied* 24, 054079, November 2025.
10. **Kazeev, Nikita**, et al. [WyckoffTransformer: Generation of Symmetric Crystals](#). ICML 2025, May 2025.
11. **Kazeev, N.**, Al-Maeni, A.R., Romanov, I. et al. [Sparse representation for machine learning the properties of defects in 2D materials](#). npj Comput Mater 9, 113 (2023).
12. Derkach, D., **Kazeev, N.**, Ratnikov, F., Ustyuzhanin, A., & Volokhova, A. (2019). [Cherenkov detectors fast simulation using neural networks](#). *Nuclear Instruments and Methods in Physics Research Section A. I'm the corresponding author*.